Catholic Junior College

JC2 Preliminary Examinations

Higher 2

CANDIDATE NAME	
CLASS	**2T**

PHYSICS 9478/02

Paper 2 Structured Questions **27 August 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class in the boxes at the top of this page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

MARK SCHEME

FOR EXAMINER'S USE	
Q1	/ 8
Q2	/ 11
Q3	/ 11
Q4	/ 12
Q5	/ 5
Q6	/ 8
Q7	/ 20
PAPER 2	/ 75

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[Turn over

DATA: **Updated for new syllabus 9478**

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left.8.99 \times 10^{9} \text{ m F}^{-1}\right)$
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} mol^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

FORMULAE:

uniformly accelerated motion	s	=	$u\,t + \frac{1}{2}a\,t^2$
	v^2	=	$u^2 + 2\,a\,s$
work done on / by a gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T / K	=	$T / {}^{\circ}C + 273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^{\,2} - x^2\right)}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$

energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0\, e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_o I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_o NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_o nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0\, e^{-\lambda t}$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 (a) Define *gravitational potential at a point*.

……

…… [1]

L1 It is the **work done per unit mass by an external force in bringing a small test mass from infinity to that point** (without a change in kinetic energy). **B1**

Examiners' comments:

- Many candidates often omitted the phrase "per unit mass", incorrectly defining potential as total work done instead of work done on a unit mass.
- Some candidates defined potential at a point as the force acting per unit mass, confusing it with gravitational field strength.
- Candidates occasionally stated that work was done by the gravitational field instead of by an external force against the field.

(b) A satellite of mass m orbits a planet of mass M and radius R_p. The radius of the orbit is R. The satellite and the planet may be considered to be point masses with their masses concentrated at their centres. They may be assumed to be isolated in space.

For the satellite in its orbit, show that

(i) its kinetic energy E_k is given by $E_k = \frac{GMm}{2R}$,

[2]

L1 The **centripetal force (F_C) on the satellite is provided by the gravitational force (F_g)**. **B1**

$F_g = F_c$

$$\frac{GMm}{R^2} = \frac{mv^2}{R}$$

$$mv^2 = \frac{GMm}{R}$$

$$\frac{1}{2}mv^2 = \frac{1}{2}\frac{GMm}{R}$$ **M1**

$$E_k = \frac{1}{2}mv^2$$

$$= \frac{GMm}{2R} (shown)$$ **A0**

Examiners' comments:

- Many candidates often started directly with algebraic manipulation without first stating the link that the gravitational force provided the necessary centripetal force.
- A few candidates attempted to derive the expression starting from $E_k = \frac{3}{2}kT$ or ideal gas expressions instead.
- Some candidates were confused with the orbit radius and planet radius in their derivation.

(ii) the ratio $\frac{\text{kinetic energy of satellite } E_k}{\text{total energy of satellite } E_T}$ is equal to -1.

[2]

L1

$E_P = -\frac{GMm}{R}$

$E_T = E_P + E_k$

$= -\frac{GMm}{R} + \frac{GMm}{2R}$ **M1**

$= -\frac{GMm}{2R}$

$\therefore \frac{E_k}{E_T} = \frac{\frac{GMm}{2R}}{-\frac{GMm}{2R}}$ **M1**

$= -1 (shown)$ **A0**

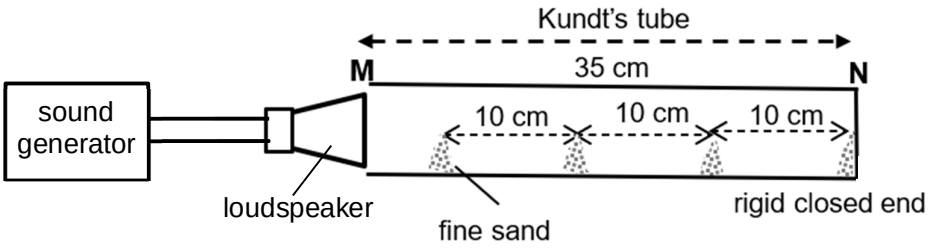
Candidates must show how they obtain E_T.

Examiners' comments:

- Many candidates often failed to write out the explicit summation $E_T = E_k + E_P$ to establish how total energy was calculated.
- Many candidates merely stated the formula for total energy without any derivation.
- Some candidates defined potential energy using *mgh* instead of the general formula $E_P = -\frac{GMm}{R}$.

	(c)	The satellite in **(b)** gradually loses energy due to small resistive forces. Suggest why many such satellites eventually 'burn up' in the Earth's atmosphere.	
		……..……..	
		……..……..	
		……..……..	
		……..……..	
		……..……..	
		……..……..	
		……	[3]
	L3	As the satellite gradually loses energy, i.e. its total energy E_T decreases (i.e. more negative), its radius of orbit R decreases. As the satellite radius of orbit R decreases, its kinetic energy E_k increases, i.e. its speed v increases as it moves nearer or into the Earth's atmosphere. The satellite's increasing speed gives rise to increase in resistive force acting on the satellite, which results in increasing rate of conversion of its energy to thermal energy, and so the satellite could eventually 'burn up' in the Earth's atmosphere.	**B1** **B1** **B1**
		Examiners' comments: • Many candidates assumed that resistive forces immediately caused the satellite to slow down, ignoring the fact that the loss of total energy caused the orbital radius to decrease. • Candidates often failed to connect the reduction in orbital radius to an actual increase in kinetic energy. • Some candidates stated that the satellite "burned up due to friction" without explaining the rate of energy conversion into thermal energy.	

[Total: 8]

2	Figure 2.1 shows a Kundt's tube of length 35 cm, containing air and fine sand, having one end open at position **M** and one closed at position **N**. A loudspeaker connected to a sound generator is placed at its open end. When the sound generator and loudspeaker is turned on, four small heaps of sand eventually formed at certain positions in the tube. Adjacent heaps of sand are at distance 10 cm apart as shown in Figure 2.1.  **Figure 2.1**	
	(a)	Explain how a stationary wave is formed in the tube.

		.....	
		.....	
		.....	
		.....	
		.....	
		.....	
		.....	[2]
	L1	**The incident sound wave is <u>reflected at the closed end</u>.** **Since the <u>incident and reflected waves</u> are of the <u>same nature, same frequency, same speed and approximately equal amplitude</u> and <u>are travelling in opposite directions along the same line (OR overlap in opposite directions)</u>, a stationary wave <u>can</u> form.** **A stationary wave <u>will</u> be formed if the tube <u>length equals to an odd number multiple of a quarter wavelength</u>.** *(as long as proper boundary conditions are stated)*	**B1** **B1**
		Examiners' comments: • A lot of students missed out on the boundary conditions or gave wrong boundary conditions. • Better answers provide detailed and full description of nature of incident and reflected waves which are traveling in opposite directions, and superposing. No marks given if students did not provide specific details on waves or did not describe the waves traveling in opposite direction.	
	(b)	Determine the wavelength of the sound produced by the loudspeaker.	
		wavelength = ... cm	[1]
	L1	The heaps of sand form at the positions of displacement nodes. Distance between successive displacement nodes = $\frac{\lambda}{2}$ $\frac{\lambda}{2} = 10\ cm \quad \Rightarrow \quad \lambda = 2 \times 10 = 20\ cm$	**A1**
		Examiners' comments: • Generally, well done. Do note however working should be shown.	

	(c)	**(i)**	The sound wave produced by the loudspeaker in Figure 2.1 has amplitude of A_o. On Figure 2.2, sketch a graph of the variation of amplitude, in terms of A_o, of the stationary sound wave with distance x along the Kundt's tube, starting from point **M** of the Kundt's tube (x = 0 cm), until its closed end at point **N** (x = 35 cm). You may assume there is no energy loss by the sound wave as it is reflected at the closed end of the tube.

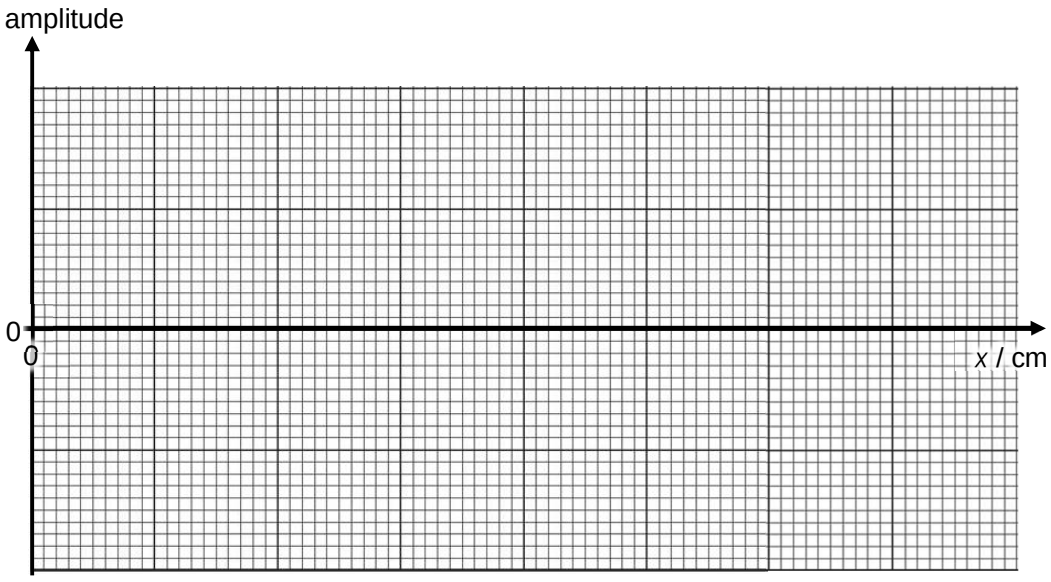
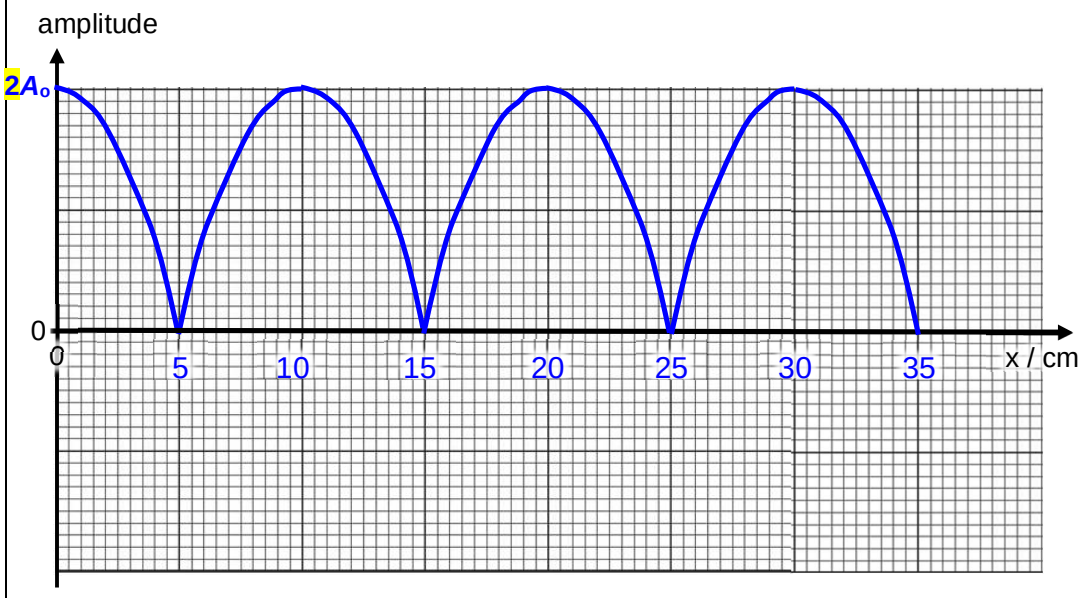


Figure 2.2 [3]

Solution:



L2 **B1- Correct shape (<u>modulus cosine</u> shape) AND 4 maximum amplitudes.** ***[This mark is for shape, correct positions of zero & maximum amplitudes is marked for in 3rd B1.]*** ***<u>Not</u>*** *cosine-**squared** shape, **unless it is Intensity** variation graph then 'squared' since Intensity proportional to Amplitude-squared.*

L3 **B1- Labelled correct amplitude of the resultant stationary wave (= $2A_o$).**

L2 **B1- Labelled correct positions of zero AND maximum amplitude on the x-axis. Good scale.**

Examiners' comments:

- Better answers give amplitude as all positive as it should. Do note that this is an amplitude versus position and not displacement versus position graph.
- A lot of students draw poorly drawn graph; incorrectly drawing linear sections or

			disjointed and wriggly drawings. Mark deducted if shape and drawing is deemed unacceptably inaccurate. Students should practice drawing graphs. • Majority of students missed the mark on the resultant amplitude.	
		(ii)	By describing the movement of air molecules in the stationary sound wave, state and explain where the air pressure varies the most.	
			……	
			……	
			……	
			……	
			……	
			……	
			……	
			……	[3]
		L2	At **displacement nodes** of the stationary wave. **Candidates must distinguish between displacement node & pressure node.* **Air molecules at either side of each displacement node are always vibrating in antiphase.** Hence, the **displacement nodes are the positions of maximum compression (OR where there is highest pressure), *as well as*, the positions of maximum rarefaction (OR where there is lowest pressure).** Thus, the **greatest variations** in pressure occur at the displacement nodes.	**B1** **B1** **B1**
		Examiners' comments: • A lot of students incorrectly state that pressure variation happens at region between displacement node and antinode. • Many wrong answers given relating maximum displacement of air particles to maximum pressure experienced. • A lot of students misread or fail to understand meaning of pressure variation versus pressure. Maximum pressure variation means where the CHANGE in pressure is the greatest. • A lot of students mistakenly just concluded based on positions where maximum pressure is felt without discussing about lowest pressure occurring at the same position. • Better answers state displacement node and wrote down corresponding positions on graph. Answers which just state positions of maximum pressure variations as N and A are not accepted.		

(iii) The maximum air pressure in the Kundt's tube is P_o.

On Figure 2.3, sketch a graph of air pressure variation, in terms of P_o, against distance x along the Kundt's tube from point **M** (x = 0) to point **N** (x = 35 cm).

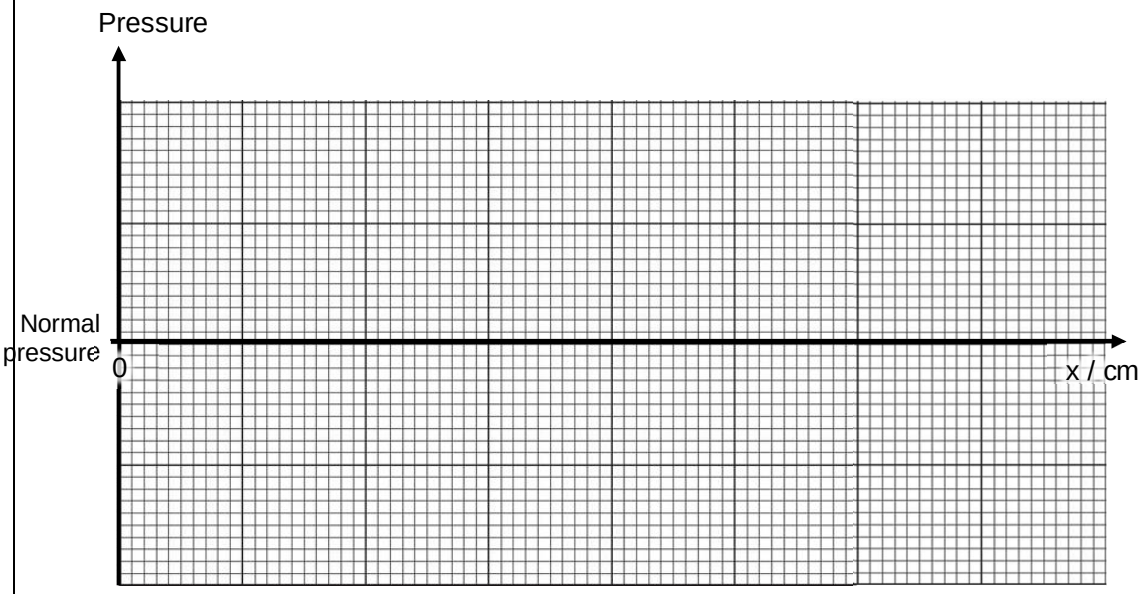
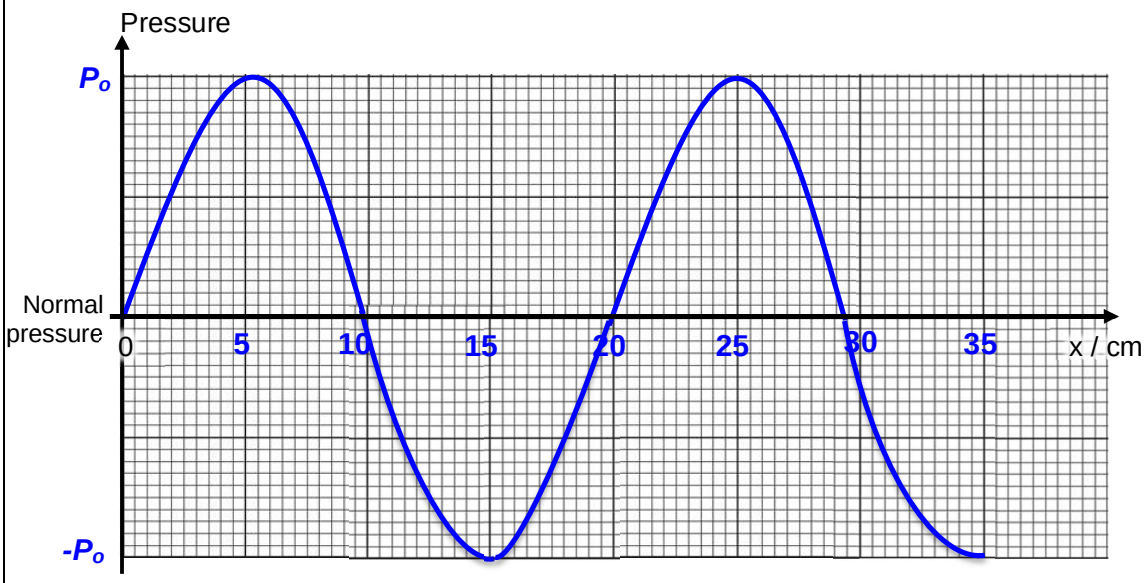


Figure 2.3 [2]

L3 Solution:

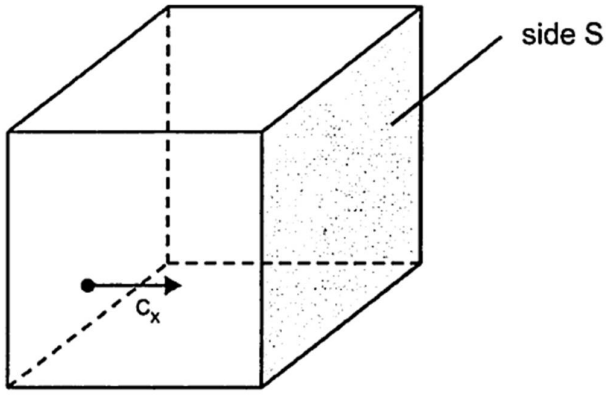


B1: Correct shape (sinusoidal; *either* positive sine *or* negative sine curve) with positive and negative P_o.
B1: Labelled correct positions of Zero AND Maximum Pressure variations along the x-axis.
Allow ECF from (c)(i).

Examiners' comments:

- Graph of pressure variation should show negative pressure.
- No marks given for incomplete graph not drawing up to x = 35 cm and mark deducted for poorly drawn (shape) graph.
- ECF given from part c(i).

[Total: 11]

3	**(a)**	The kinetic theory of gases is based on some simplifying assumptions. For example, the molecules of the gas are assumed to behave as hard elastic identical spheres. State the basic assumption about the molecules of an ideal gas based on		
		(i)	the nature of their movement,	
			……	
			……	[1]
		L1	Accept either one: **Random motion** OR **Constant velocity between collisions**	**B1**
			Examiners' comments: • Generally well done by most students. • Answers accepted are on nature of motion (or nature of movement in between collisions). Description on type of collisions not accepted.	
		(ii)	their volume.	
			……	
			……	
			……	[2]
		L1	**Total volume of molecules is negligible compared to volume of containing vessel /container.**	**B1** **B1**
			Examiners' comments: • Some students mistakenly write about size of gas molecules instead of volume, which is not accepted as size can means other dimensions such as diameter. • Small number did not make comparison to volume of container.	
	(b)	A cube of volume V contains N molecules of an ideal gas. Each molecule has a component c_x of velocity normal to one side S of the cube, as illustrated in Figure 3.1.  **Figure 3.1**		

The pressure p of the gas due to the component c_x of velocity is given by the expression

$$pV = Nm\langle c_x^2 \rangle$$

where m is the mass of a molecule and $\langle c_x^2 \rangle$ is the mean-square-speed of the molecules in the direction normal to side S of the cube in Figure 3.1.

Show and explain how the above expression leads to the relation

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

where $\langle c^2 \rangle$ is the mean-square-speed of the molecules.

[3]

L3 *Explain how the kinetic theory of gases is applied in your answer.*

Based on the assumptions of the kinetic theory of gases,

1. **all gases consist of a very large number *N* of identical particles, and,** **B1**

2. **all the particles move randomly** (at varying speeds and move in all directions with equal probability**). Implying that mean square speed component in any one of the three mutually perpendicular directions (*x*, *y* and *z*) is the same.** **B1**

i.e. $\langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle$

$\langle c^2 \rangle = \langle c_x^2 \rangle + \langle c_y^2 \rangle + \langle c_z^2 \rangle$ **B1**

Therefore,

$\langle c_x^2 \rangle = \frac{1}{3}\langle c^2 \rangle$

Hence, $pV = Nm\langle c_x^2 \rangle$ **leads to** $pV = Nm.\frac{1}{3}\langle c^2 \rangle = \frac{1}{3}Nm\langle c^2 \rangle$ A0

Examiners' comments:

- Majority of students miss the first B1 mark.

- Some students wrote the wrong notation for mean squared speed, and did not define the symbols they used.
- No mark given if students 'work backwards' starting from derived expression.
- A large number of students did not read the question or understood what is required for this question which is to derive expression based on earlier equation given. A large number of students incorrectly attempted to derive from other equations of ideal gas or Newton's Laws.

(c) Neon gas behaves as an ideal gas.

A mass of 20.0 g of neon gas occupies a volume of 2.24×10^4 cm^3 at a pressure of 1.01×10^5 Pa and a temperature of 273 K.

Deduce the root-mean-square (r.m.s.) speed of neon atoms at a temperature of

(i) 273 K,

r.m.s. speed = ……m s^{-1} [3]

L2 Note that **total mass** of gas = ***Nm*** = **20.0 g**

$pV = \frac{1}{3}Nm\langle c^2\rangle$

$(1.01 \times 10^5)(2.24 \times 10^4 \times 10^{-6}) = \frac{1}{3}(20.0 \times 10^{-3})\langle c^2\rangle$ **M1**

$\langle c^2\rangle = 3.3936 \times 10^5$ **C1**

$\sqrt{\langle c^2\rangle} = \sqrt{3.3936 \times 10^5} = 582.55 = 583$ m s^{-1} **A1**

Examiners' comments:

- Not well done. A large number of students mistakenly took 20 g as molar mass or mass of one molecule. It is the total mass.
- If students had used ideal gas expression (3/2NKT or 3/2nRT) to equate to K.E expressions, they have to calculate N or n, which some mistakenly missed out.
- A large number of students substituted wrong values or made power of 10 errors.
- A number of students left final answer at mean squared speed instead of root mean square speed.

		(ii)	27 ˚C.	
			r.m.s. speed = ……..……..…..m s^{-1}	[2]
		L2	**$<c^2> \propto T$** $\frac{\langle c^2 \rangle_{27°C}}{\langle c^2 \rangle_{273\ K}} = \frac{27 + 273.15}{273}$ $\frac{\langle c^2 \rangle_{27°C}}{3.3936 \times 10^5} = \frac{27 + 273.15}{273}$ **M1** $\langle c^2 \rangle_{27°C} = 3.7311 \times 10^5$ $\sqrt{\langle c^2 \rangle_{27°C}} = 610.83 =$ **611 m s^{-1}** **A1**	
			Examiners' comments: • A number of students fail to realise that PV of gas would change with change in temperature. Instead students mistakenly assumed amount of gas to change and calculated a new amount of gas to substitute into calculation. • ECF given if students shows correct method showing $<c^2> \propto T$	

[Total: 11]

4	**(a)**	By reference to energy, explain what is meant by *electromotive force* (e.m.f.).	
		……....	
		……....	
		……	[1]
	L1	e.m.f. is the **amount of energy transferred per unit charge from some other form of energy into electrical energy when charge is moved round a complete circuit.**	**B1**
		Examiners' comments: • It is important to state emf as the "energy per unit charge". This is one phrase and is a nuance different from "driving a unit charge around a complete circuit". • Many students were vague about the definition of emf. Emf is a physical	

quantity and must be linked to a physical quantity e.g. energy, force etc. that has units. These students loosely mention it as a “conversion of energy”, and the action of “conversion” is not a physical quantity.

(b) A ‘potato cell’ is formed by inserting a copper plate and a zinc plate into a potato. The circuit shown in Figure 4.1 is used in an investigation to determine the e.m.f. and internal resistance of the potato cell.

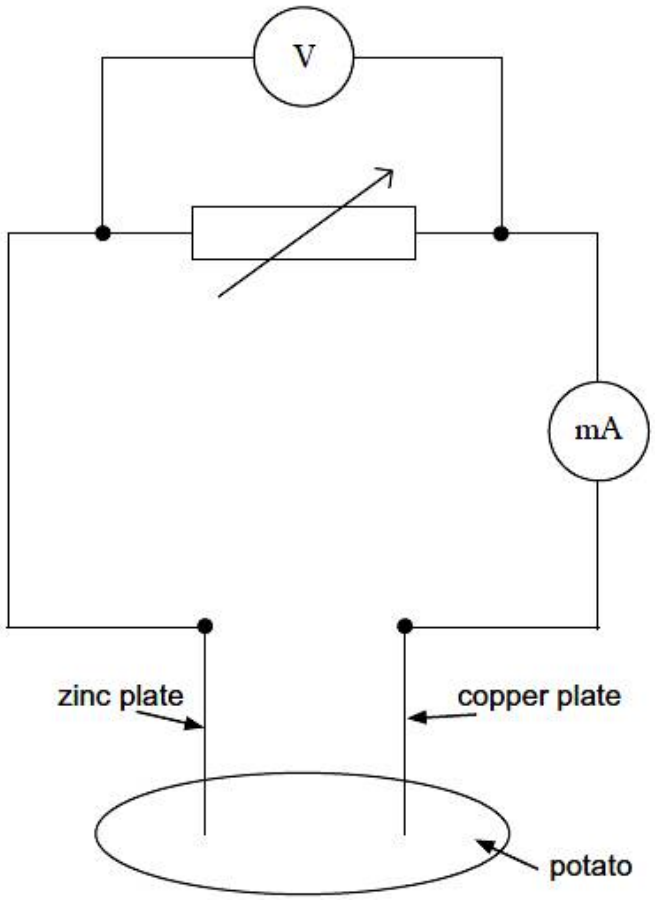


Figure 4.1

The plotted points × on Figure 4.2 show the data for current and voltage that were obtained by varying the resistance of the variable resistor. The line of best fit was then drawn through the plotted points.

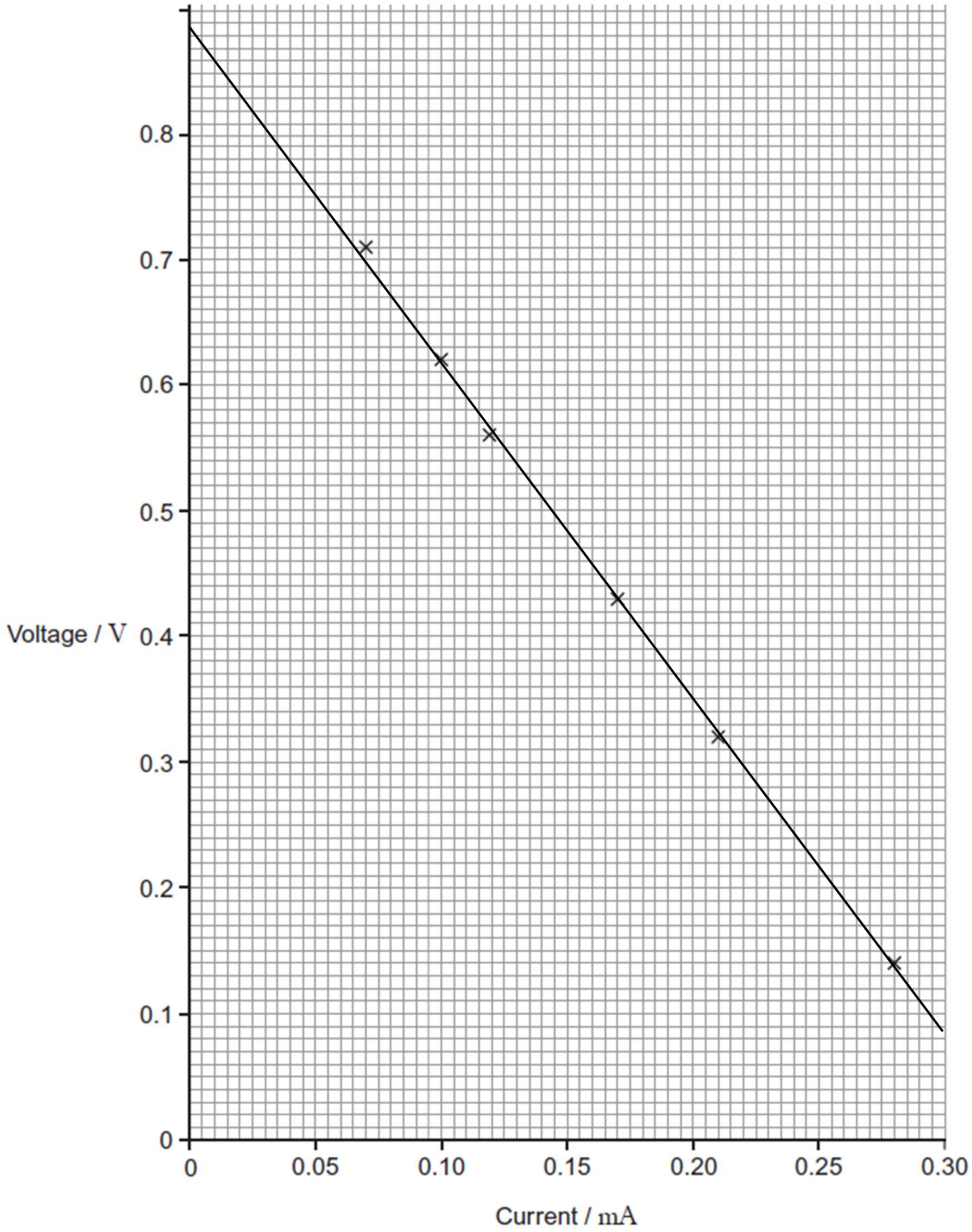
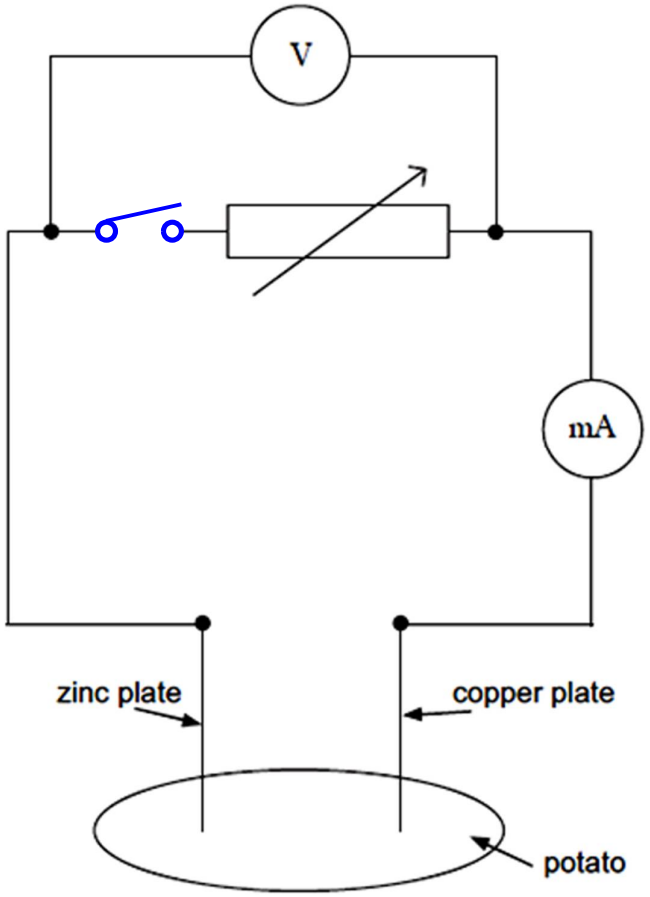


Figure 4.2

		(i)	State the e.m.f. of the potato cell.	
			e.m.f. = ……V	[1]
		L2	The graph in Figure 4.2 is described by the mathematical equation: $V = E - Ir$ When $I = 0$, $V = E$. Hence, the ***V*-intercept is the e.m.f.** → **e.m.f. = <u>0.885 V</u>** **point must **be read off accurately to <u>half</u> the smallest division** on the graph, hence 0.885 V <u>not</u> 0.89 V.*	**B1**
			Examiners' comments:	

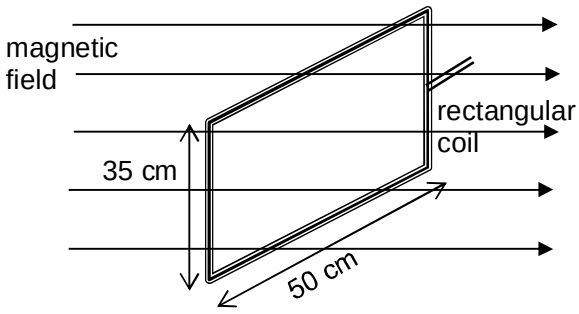
			• Many students did not read to half the smallest square on the graph.	
		(ii)	Determine the internal resistance of the potato cell.	
			internal resistance = ……Ω	[3]
		L2	The graph in Figure 4.2 is described by the mathematical equation: $V = E - Ir$ **Internal resistance = – gradient** $= -\left[\frac{0.885-0.310}{(0-0.215)x10^{-3}}\right]$ = 2674.4 = **2670 Ω** ***C1 mark*** *– Evidence of equating internal resistance to negative of the gradient.* ***M1 mark*** *– Coordinates used must be separated by more than half the length of the line of best fit, and, correct reading off of coordinates. **If there is power of ten error, deduct under A1 mark, award M1 mark.* ***A1 mark*** *– Accept answers in the range 2600 – 2800 Ω, provided M1 mark scored.*	**C1** **M1** **A1**
			Examiners' comments: • Sufficient credit is given to students who are able to use V = E – Ir, without ambiguities. • Many students are not clear of V and E in the abovementioned equation. • It is best to distinguish between external load and internal resistance using R and r respectively.	
		(iii)	Determine the voltage across the variable resistor when the power delivered to it is maximum. Explain your working.	

			voltage = ……V	[3]
		L2	**Power output is maximum when** **variable resistance *R* = internal resistance of the potato cell *r***	**B1**
			Using potential divider principle, **voltage across the variable resistance = 0.885 / 2**	**M1**
			= 0.4425 **= 0.443 V**	**A1**
			Examiners' comments: • Many students could not recall maximum power theorem to solve this question. • Credit can only be given when students show clear understanding, regardless of the final value.	
		(iv)	Explain why the voltages of the plotted points × on Figure 4.2 are always less than the e.m.f. of the potato cell.	
			……	
			……	
			……	
			……	
			……	
			……	
			……	
			……	[3]
		L2	The **variable resistor's (maximum) resistance is not infinite. Current flows through the potato cell** regardless of how the variable resistor's resistance is varied **(OR, current is never zero)**, **causing a** (non-zero) **potential difference to exist across the internal resistance of the cell.**	**B1**
			Consequently, the terminal potential difference of the cell is always less than its e.m.f.	**B1**
			The **voltmeter reads the terminal potential difference** of the potato cell (not the e.m.f.).	**B1**
			Examiners' comments: • Many students did not mention the voltages of the plotted points on Figure 4.2 is the terminal potential difference. • Many students also did not understand the meaning of terminal potential difference and its presence in the equation V = E – Ir. • It is often not clear what the students refer to that is always less than emf. • Students did not realise the current flowing through the internal resistance is the key point. • Students wrongly bring in energy loss, which introduce more ambiguities and fail to help them explain their answers.	

		(v)	In the space below, draw a circuit diagram to suggest a modification to the circuit in Figure 4.1 which allows the e.m.f. value of the potato cell to be read directly from the voltmeter.	
				[1]
	L2	**Solution:**  • **Add in a switch in series with the rheostat.** • **Switch must be in between the 2 terminals of the voltmeter.**		**B1**
		Examiners' comments: • As current flowing through the internal resistance is the key point, it is only		

		logical that the correct answer is to break the circuit such that there is no current flow.	

[Total: 12]

5	**(a)**	A simple a.c. generator which consists of a rectangular coil with 40 turns rotates at 50 revolutions per second in a uniform magnetic field of flux density 50 mT. The coil is 50 cm long and 35 cm wide, as shown in Figure 5.1. An electromotive force (e.m.f.) is induced in the coil.  **Figure 5.1** The maximum induced e.m.f. E_o in the coil is given by the expression E_o = maximum magnetic flux linkage × angular speed of the coil Use this expression to calculate the maximum e.m.f. induced E_o.	
		E_o = ……V	[2]
	L1	$E_o = \Phi_{max} \times \omega$ $= NBA \times 2\pi f$ = **(40) (50 x 10^{-3}) (50 x 10^{-2})(35 x 10^{-2}) × (2π)(50)** = 109.96 = **110 V (Accept 2 s.f. or 3 s.f.)**	**M1** **A1**
		Examiners' comments: • A sizeable number of students were not able to substitute the correct values.	
	(b)	The output from the generator in part **(a)** is connected to the input of a transformer. The transformer has 40 turns on its primary coil and 10 turns on its secondary coil. The transformer may be considered to be ideal. The secondary coil is connected in series to a resistor of resistance 100 Ω and a diode, as shown in Figure 5.2.	

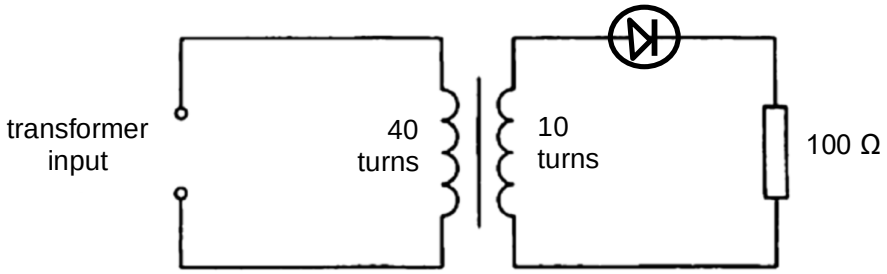


Figure 5.2

Calculate the root-mean-square voltage across the 100 Ω resistor.

root-mean-square voltage = ……V [3]

L3

Let V_0 be the **peak** voltage across the resistor.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\frac{\boldsymbol{V_0}}{\mathbf{109.96}} = \frac{\mathbf{10}}{\mathbf{40}}$$ **M1**

$V_0 = 27.49$ V

Since the voltage across the resistor is **half-wave rectified**,

$\langle V^2 \rangle = \frac{V_o^2}{4}$

$\therefore \sqrt{\langle V^2 \rangle} = \frac{V_o}{2} = \frac{\mathbf{27.49}}{\mathbf{2}}$ **M1**

= 13.745 = **14 V (Accept 2 s.f. or 3 s.f.)** **A1**

Examiners' comments:

- A handful of students were not able to understand since the input voltage is the peak voltage on the primary arm of the transformer, the output voltage at the secondary arm is also the peak voltage.
- Others were not able to apply the correct equation for half-wave rectification.

[Total: 5]

6 An electron is confined within a one-dimensional box of width L.

ψ represents the sinusoidal wavefunction of the electron. The graph of $|\psi|^2$ against distance x from one wall of the box is shown in Figure 6.1.

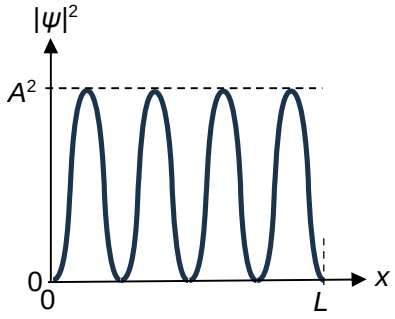


Figure 6.1

(a) (i) State what is represented by $|\psi|^2$.

……

…… [1]

L1 **Probability density (OR Probability per unit length) of finding the position of the electron.** **B1**

Examiners' comments:
- Many candidates often omitted the per-unit-length aspect, defining $|\psi|^2$ as "probability" instead.
- Some candidates confused $|\psi|^2$ with the wavefunction ψ itself, describing it as the amplitude or displacement of the wave.
- Some candidates provided generic statements without referencing the electron.

(ii) The graph of $|\psi|^2$ against x in Figure 6.1 has an amplitude of A^2.

Derive an expression for A in terms of L.

A = …… [2]

L1 Total Area under the graph = Total Probability = 1

$\int_0^L |\psi|^2 dx = 1$

$\frac{A^2}{2} L = 1$ **M1**

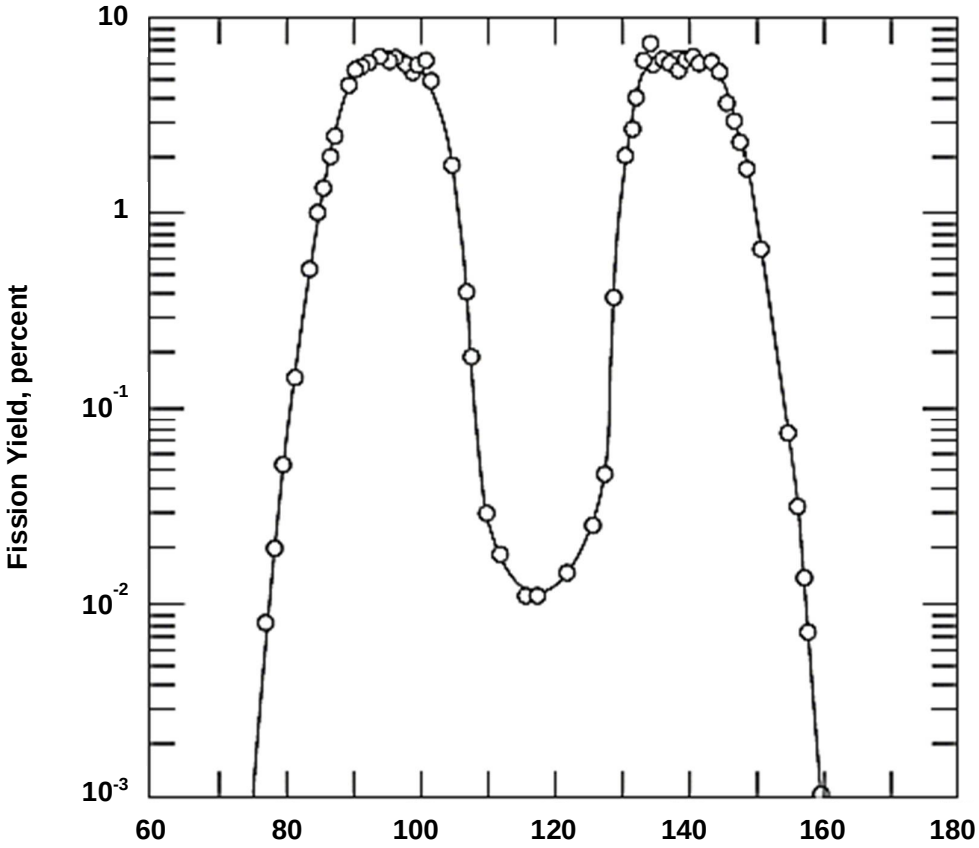
			$A = \sqrt{\frac{2}{L}}$	**A1**
			Examiners' comments: • Many candidates often failed to identify the significance of the total area under the $\|\psi\|^2$ graph. • Some candidates incorrectly assumed that the peak value A^2 itself must be equal to 1. • A few candidates occasionally substituted random numerical values for L.	
	(b)	The electron with the $\|\psi\|^2$ graph shown in Figure 6.1 is in an excited quantum state n in a box where L = 0.10 nm.		
		(i)	State the quantum state n of the trapped electron shown in Figure 6.1.	
			n = ……	[1]
		L2	**n = 4**	**B1**
			Examiners' comments: • Many candidates confused the quantum state number n with the number of internal nodes rather than anti-nodes. • Some candidates gave non-integer values or fractions for n. • A few candidates confused the quantum number n with the box length L.	
		(ii)	Calculate the momentum of the electron at this quantum state.	
			momentum =…… N s	[2]
		L2	**Wavelength, λ = L / 2 = 0.10 / 2 = 0.050 nm = 0.050 x 10^{-9} m** $p = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{0.050\times10^{-9}}$ = 1.326 x 10^{-23} **= 1.3 x 10^{-23} N s**	**M1** **A1**
			Examiners' comments: • Many candidates forgot to convert the units of the box width L from nanometers to meters, leading to power-of-ten errors. • Some responses misquoted the value of Planck constant h. • Some candidates attempted to use $p = mv$ to solve without identifying the value of v.	

		(iii)	Calculate the energy, in eV, of the electron at this quantum state.	
			energy = …….. eV	[2]
		L1	$E_n = \frac{n^2h^2}{8mL^2}$ $= \frac{(4)^2(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(0.10\times10^{-9})^2}$ = 9.65025 x 10^{-17}J **= 603 eV** ***Allow ECF of value of n in part (b)(i).*** **OR use $E_n = \frac{p^2}{2m}$ where *p* is found in (b)(ii).**	**M1** **A1**
			Examiners' comments: • Some candidates misread and used wrong formulas for energy levels. • Some candidates used an incorrect value for the mass of an electron, substituting the mass of proton instead. • A few candidates forgot to convert the units for the calculated energy from joules to electron volts.	

[Total: 8]

7	When a very strong earthquake of magnitude 9.0 hit Japan on 11 March 2011, the accompanying tsunami caused the devastation of the Fukushima Dai-ichi Nuclear Power Plant (Fk-1). A large amount of radioactive material leaked out of Fk-1 as a result, mainly through cracks made by the quake on some reactors, explosions, intentional vents to relieve pressure, and leakage of cooling water that was contaminated with the melted fuel rod debris. Fk-1 was capable of generating up to 4.7 gigawatts of power through neutron-induced fissions of Uranium-235 (^{235}U) nuclei. Fission resulting from neutron absorption is called *induced* fission. Some nuclides can also undergo *spontaneous* fission, but that is quite rare. Nuclear fission is a decay process in which an unstable nucleus splits into two fragments of comparable mass, although nearly equal mass is unlikely. Over 100 different nuclides, representing more than 20 different elements, have been found among the fission products. Figure 7.1 shows the distribution of mass numbers for fission fragments from the fission of

^{235}U. Fission yield refers to the percentage of a nuclide of a fission fragment produced per fission. Two or three free neutrons usually appear along with the fission fragments.



Source: https://www.quora.com/Why-is-promethium-produced-from-uranium-235

Figure 7.1

The induced fission of ^{235}U may be represented by a nuclear equation of the form

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{a}_{b}P + {}^{c}_{d}Q + (2 \text{ or } 3)\,{}^{1}_{0}n + \text{energy}$$

where P represents the lower mass number fission fragment, and Q represents the higher mass number fission fragment.

One day after the earthquake, elevated levels of Iodine-131, Cesium-134 and Cesium-137, which are common fission fragments of Uranium-235, were detected near the nuclear plant site. Figure 7.2 shows some general information of the three fission fragments.

Isotope	Symbol	Half-life
Iodine-131	$^{131}_{53}I$	8.0197 days
Cesium-134	$^{134}_{55}Cs$	2.065 years
Cesium-137	$^{137}_{55}Cs$	30.17 years

Figure 7.2

	(a)	Explain the difference between *induced* and *spontaneous* fission.	
		……...	
		……...	
		……	[1]
	L1	Unlike spontaneous fission, **induced fission requires** (initial) **neutron absorption to occur**. *Candidates need to answer this question by reading the passage in context.*	**B1**
		Examiners' comments: • Some candidates used imprecise terms like neutron 'collision' / 'bombardment' instead of 'absorption'. Also, although the spelling is given in the passage, many candidates mis-spelled 'absorption' as 'absorbtion'. Candidates to note that Cambridge can deduct marks for wrong spelling of key words. • Some candidates do not know the difference between 'random' and 'spontaneous'. • Some candidates did not read the passage and gave a generic answer like "Induced fission is caused by an external stimulus / external agent." Some candidates mentioned that an "external net force is needed".	
	(b)	Suggest why the *percentage fission yield* in Figure 7.1 is shown on a logarithmic scale.	
		……...	
		……...	
		……	[1]
	L3	Percentage fission yield varies over a wide range of many orders of magnitude. OR Logarithmic scale compresses the scale to accommodate the very wide range of percentage fission yield values. OR To enable plots on the graph to be well spaced while representing the very wide range of percentage fission yield values.	**B1**
		Examiners' comments: • Many correct answers and also many incorrect answers. • Candidates with incorrect answers talked about extremely low values, or, extremely high values, rather than about the wide range. • Some candidates said that fission yield as an exponential function, however this is not a natural logarithm graph nor a linear graph.	
	(c)	Using Figure 7.1, state the two most common ranges of mass numbers of the fission fragments (P and Q) of ^{235}U.	
		most common range of mass numbers of P: ……	[1]
		most common range of mass numbers of Q: ……	[1]
	L1	From Figure 7.1, Most common range of mass numbers of **P**: **90 to 100 (Accept lower boundary value 88 to 90, and, upper boundary value 100 to 102.)** Most common range of mass numbers of **Q**: **132 to 142 (Accept lower boundary value 132 to 134, and, upper boundary value 142 to 145.)**	**A1** **A1**

Note that Question has already defined P as the fragment with *lower* mass. So answers for P and Q cannot swop.
Marking points:
- Correct reading of the ranges of values.
- Answers for P and Q not swopped.

Examiners' comments:
- Several candidates did not read off the A-coordinates accurately, e.g. many drew lines that were slanted when reading off from the A-axis, or obviously used free hand rather than a ruler to read off accurately.
- Some candidates gave single mass number instead of a *range* of mass numbers.

(d) (i) Uranium-235 undergoes nuclear fission to produce Rubidium-97 and Cesium-137.

Write down the nuclear equation to represent this reaction. Use the chemical symbol Rb to represent Rubidium.

…… [2]

L2 $^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{97}_{37}Rb + {}^{137}_{55}Cs + 2\,{}^{1}_{0}n$

- Correct value of proton number of Rb (= 37). **A1**
- Deduced that 2 neutrons are emitted. **A1**
- Deduct 1 mark for other mistakes e.g. used "=" instead of "→"; did not include $^{1}_{0}n$ on the LHS of the equation; wrong notations e.g. $n\,^{1}_{0}$ or $^{2}_{0}n$ instead of $2\,^{1}_{0}n$

Note: General equation is given in the question, hence no mark purely for writing the equation.

Examiners' comments:
- Many candidates answered this successfully.
- However, many other candidates did not notice that the general equation and the nuclide notation for Cs-137 are both given in the question on pages 20 & 21.

(ii) With reference to Figure 7.1, state the percentage fission yield of Cesium-137 and Rubidium-97.

percentage yield of Cesium-137= ……%

percentage yield of Rubidium-97 = ……% [1]

L1 6% for both. (Accept also 6.5%.) **A1**
Mark for correct method (clear indication of points read on Figure 7.1).
Accept also if answers given are 6.5% for one, and 6% for the other.

Further interesting note on Figure 7.1:
By conservation of mass number before and after the fission, it is necessary that the two fission fragments are either read off the two inner parts of the "M" shaped curve in Figure 7.1, or the two outer parts.

Examiners' comments:
- Common incorrect read-offs are: 5, 7, 8, 60.

		(iii)	Suggest why both the answers in **(d)(ii)** are similar.	
			……..……	
			……..……	
			……..	[1]
		L3	Percentage yields of both should be similar since **they are produced from the same (unique) fission process**. So the probabilities of both of them forming are equal/same. *Do NOT accept:* "In the nuclear equation, Cesium-137 and Rubidium-97 are produced in a 1:1 ratio." ➔ *This is NOT accepted as an answer because: Firstly, it is rephrasing the question which already specifies that the answers are 'similar'. Secondly, it does not address the meaning of 'percentage yield' which the values in (d)(ii) refer to. In paragraph 2 of the given text, the concept of percentage yield comes in because U-235 can undergo many different fission processes, i.e. it does not necessarily split into Cs and Rb. There is also a need to justify that Cs and Rb are produced in the same unique fission process, i.e. no other fission possibilities of U-235 can yield either one of them without the other.* *Do NOT accept:* "due to conservation of mass number", *because question asks for reasons for similarity in the percentage fission yield, not about deducing 1 product from the other given product. Moreover, there are neutrons not accounted for by this explanation.*	**B1**
		Examiners' comments: • Few candidates answered this question successfully. • Many candidates said both are products of the fission of U-235, or, that each fission of U-235 produces Cs-137 and Rb-97 in a 1:1 ratio. However, the passage informed that U-235 can undergo many different fission processes, i.e. does not necessarily split into Cs-137 and Rb-37!		
		(iv)	Determine the **two** possible mass numbers of the other fission fragment produced if one of the fragments has a mass number of 81 in the nuclear fission of Uranium-235.	
			mass number = ……. or …….	[2]
		L2	Using ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{81}_{b}P + {}^{c}_{d}Q + (2 \text{ or } 3){}^{1}_{0}n + \text{energy}$ By conservation of mass number, **235 + 1 = 81 + c + (2 or 3)** **c = 153 or 152**	**A2**
		Examiners' comments: • Many correct answers. • Candidates to note that the general nuclear equation is given on page 20!		

	(e)	Determine which isotope in Figure 7.2 will be about 47% as radioactive after 33 years. Show your reasoning.	
		isotope: ……	[2]
	L2	Method 1: **47% ≈ 50%, hence** **33 years is close to the half-life of the isotope.** From Figure 7.2, the isotope with a half-life close to 33 years is **Cesium-137**. Method 2: **Activity at time t, $A = A_o e^{-\lambda t}$ where $\lambda = \ln 2 / t_{1/2}$** **"47% as radioactive" → $A / A_o = 0.47$** **$0.47 = e^{-\lambda(33 years)}$** $\lambda = 0.022879$ years^{-1} $\ln 2 / t_{1/2} = 0.022879$ years^{-1} **$t_{1/2}$ = ~30 years** **Answer: Cesium-137**	**M1** **A1** **[M1]** **[A1]**
		Examiners' comments: • Many candidates answered this question successfully. • A1 mark is not awarded if M1 mark not scored, i.e. reasoning not seen or unclear. Candidates are reminded to explain/show their reasoning clearly! E.g. Several candidates did not show full working involving stating the full equation first followed by substitution of values, and/or, showed a string of numbers without statements of what those numbers mean. • Stating that 33 years is close to the half-life of Cs-137 is an incomplete answer. Reference must be made to "47%" being approximately close to "50%" as well.	
	(f)	The release of Iodine-131, Cesium-134 and Cesium-137 into the air, ground and seawater were monitored closely by the Japanese authority because they readily decay by emitting beta-particles and gamma-rays. For example, Cesium-137 decays into Barium-137 (Ba) through beta-emission. 95% of Cesium-137 decays to a metastable excited nuclear energy state (Barium-137m) and is responsible for all emission of gamma-rays, while the other 5% directly decays to the ground state (Barium-137) of the nucleus. Figure 7.4 illustrates the decay scheme of Cesium-137.	

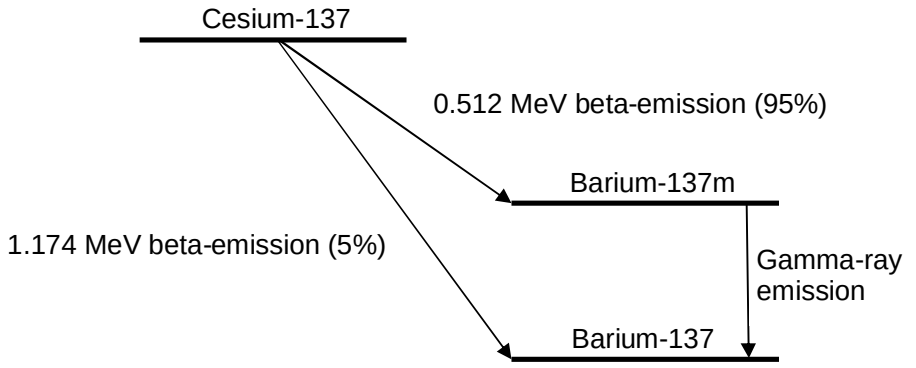


Figure 7.4

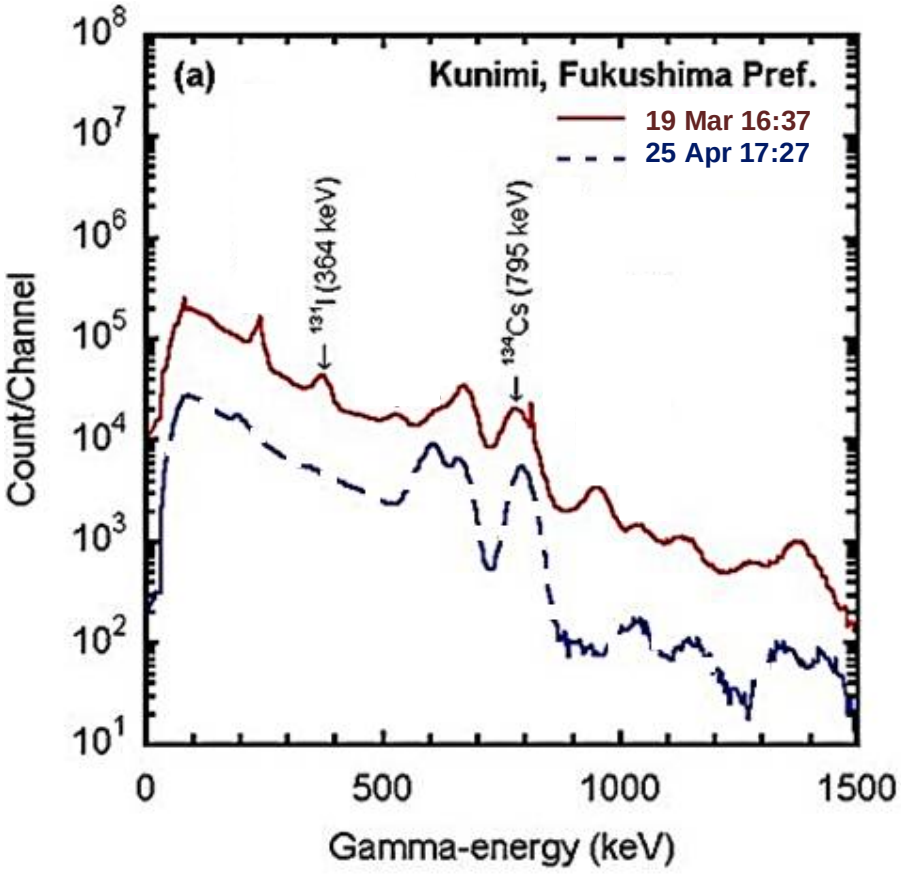
Determine the photon energy of the gamma-rays emitted during Cesium-137 decay.

energy = ……MeV [2]

L1 Energy of gamma-ray photon
= **1.174 – 0.512** **M1**
= **0.662 MeV** **A1**

Examiners' comments:
- Many correct answers.
- However, many other candidates multiplied the values of 1.174 MeV and 0.512 MeV by 5% and 95% respectively. Candidates to note that **one** gamma-ray photon is emitted by the de-excitation of **one** unstable nucleus.

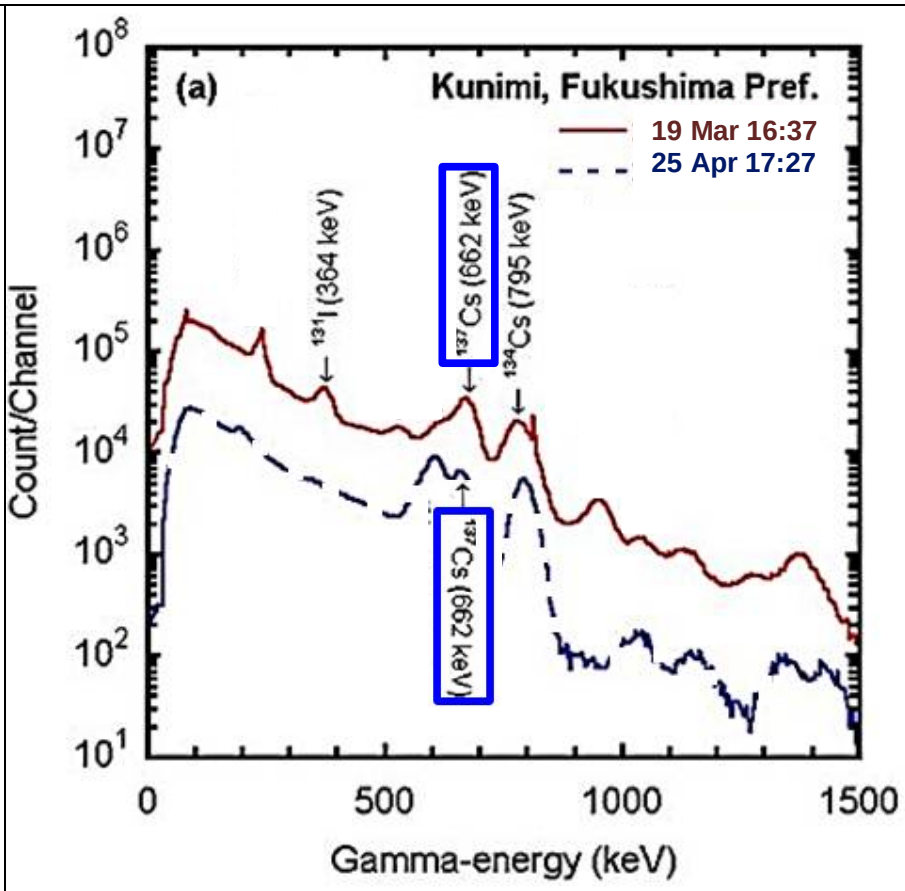
(g) Gamma spectroscopy is a technique used by scientists to identify specific radioactive products from the many possible products that can be produced from the nuclear fission of Uranium-235. Figure 7.5 shows the gamma spectra measured at Kunimi, Fukushima Prefecture on both 19 March 2011 and 25 April 2011. Photon peaks generated by Tellurium-132, Iodine-131, -132, and Cesium-134, -136 and -137 were observed in the spectra given in Figure 7.5. The photon peaks corresponding to Iodine-131 and Cesium-134 are indicated in Figure 7.5.



Source: https://www.nature.com/articles/srep00087 (modified)

Figure 7.5

		(i)	Using your answer in **(f)**, draw an arrow on Figure 7.5 to show the position of Cesium-137 in the gamma spectra. Label this arrow Cs-137.	[1]
		L2	**0.662 MeV = 662 keV** ***Accept indication on either one of the solid or dotted line graphs.*** ***Allow ECF according to candidate's answer in (f). Check for <u>correct units conversion from MeV to keV.</u>***	**A1**



Examiners' comments:

- Most candidates scored E.C.F. mark from part (f) as they converted units correctly from MeV to keV.
- Although question stated photon 'peaks' are generated by Cs-137, some candidates positioned the arrow slightly off the peak. While B.O.D. is given is scripts where arrow is not too far off, candidates should improve their precision of graph work and/or read question more carefully.

(ii) Suggest why no photon peak of Iodine-131 was observed on 25 April 2011.

……

……

…… [1]

L3 As seen from Figure 7.2, Iodine-131 has a short half-life of about 8 days, hence **a month after 19 March, on 25 April, about four half-lives had passed, hence most of the iodine nuclei would have already decayed**. **B1**

OR

Half life of Iodine-131 is about 8 days, so **by 25 April, yield of Iodine-131 is only 4.42% of that on 19 March which is significantly low.**

[**About 36 days passed** → $\frac{t}{t_{\frac{1}{2}}} = \frac{36}{8} = 4.5\ half\ lives\ passed$ →

$\frac{A}{A_o} = \left(\frac{1}{2}\right)^{4.5} = 0.0442 =$ **4.42%**]

Examiners' comments:

- Many candidates did not **use data to substantiate their conclusion that most** of the I-131 nuclei would have already decayed, e.g. many candidates simply re-stated the half-life of I-131 in Fig. 7.2 without using it to evaluate the number of half-lives elapsed or the percentage of I-131 remaining.

(h) Compare the three types of radioactive decay: alpha-decay, beta-decay and gamma-decay. Explain which of the three poses the greatest health risk if the source is

(i) outside the body,

……

……

……

……

…… [2]

L2 **Gamma-decay. Because gamma-ray is the most penetrating hence penetrate most deeply into the body and reach living tissues and vital organs.** **B1**

Alpha-particle cannot penetrate the skin. Beta-particle can penetrate the skin but cannot penetrate as deeply to reach vital organs. **B1**

Must link to health risk for the 2nd mark.
No mark for simply stating Gamma-decay without the correct reason.

Comparative language is required e.g. "most penetrating"/ "highest penetrating ability".
Elaboration of what kind of health problems is not required for part i, since the link for the 2nd mark should be made to the implication of "highest penetrating ability" to health → greatest potential of reaching and affected more living tissues in the body.

(All 3 types of radiations damage cells via ionization. The danger of this ionizing ability is elaborated in part ii answer. Focus on part i is on the penetration ability.)

Examiners' comments:

- Candidates are reminded to contextualise their answer to the question which focuses on 'health risk'. Many candidates talked about alpha-particles being stopped by thin sheet of paper, beta-particles stopped by aluminium sheet, gamma-ray stopped by lead, instead of contextualizing to human skin and body.
- Several candidates wrongly stated that beta-particles cannot penetrate the human skin.
- Candidates who merely mentioned that gamma-ray has the greatest penetrating power but did not link to health risk in comparison to alpha-particles and beta-particles did not obtain full credit.
- When comparing between gamma-ray and beta-particles, some candidates thought that since gamma-ray is the least ionizing it does not pose health risk, forgetting that 'least' does not mean 'no significant' risk. Moreover, with prolonged exposure over time, the risk can be significant. Beta-particles on the other hand cannot penetrate deeply to reach vital organs.

		(ii)	ingested and entered the body.	
			……	
			……	
			……	
			……	
			……	[2]
		L2	**Alpha-decay. Because alpha-particle is the <u>most</u> ionizing** **hence produces the <u>most</u> number of free radicals per unit distance traveled inside the body / causes the <u>most</u> extensive mutation of cells or DNA / <u>highest</u> cancer risk.** ***Must link to health risk for the 2nd mark.*** ***No mark for simply stating alpha-decay without the correct reason.*** ****Knowledge of health risks is in the A-level syllabus.*** ***Comparative language is required e.g. "most ionizing"/ "highest ionising ability".*** ***Elaboration of what kind of health problems <u>is required</u> for part ii, since the link for the 2nd mark should be made to the implication of "highest <u>ionising</u> ability" to health → e.g. production free radicals, mutation of cells or DNA, cancer risk.*** ***Repetitive answers are not accepted e.g. can ionize cells → repeating "ionizing ability", need to elaborate on health impact.***	**B1** **B1**
		Examiners' comments: • Several candidates wrote answers that seem to imply that *only* alpha-particles can cause cell damage. Candidates need to **note that all 3 types of radiations are ionizing and hence all can cause cell mutation to a certain extent. E.g. "Alpha-particles are the most ionizing type of radiation *and can cause* cell mutation and cancer." → Such an answer is lacking as it does not acknowledge that beta-particles and gamma-ray can cause cell mutation and cancer too, but to lesser extent for the same amount of exposure.**		

[Total: 20]

END OF PAPER 2

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